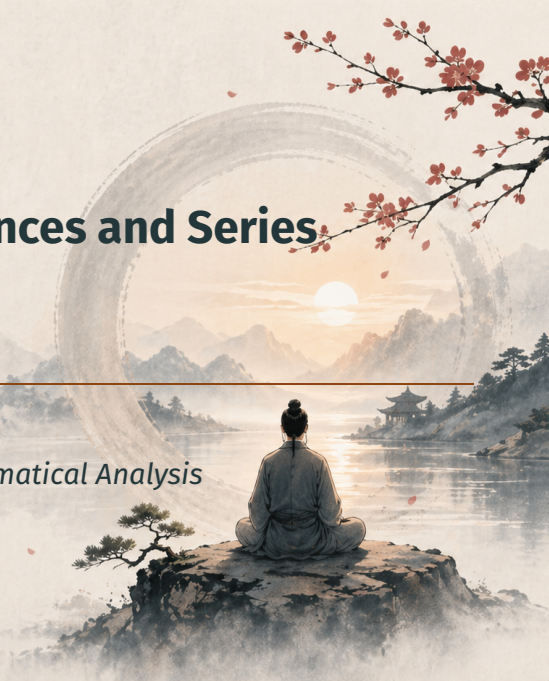


Chapter 3: Numerical Sequences and Series

Section: Some Special Sequences

Based on Walter Rudin's *Principles of Mathematical Analysis*



Outline

Motivation

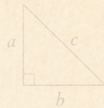
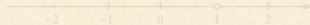
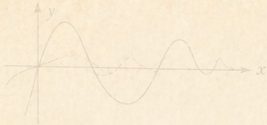
The Squeeze Principle

The Binomial Theorem

Five Special Limits

Proofs of the Five Limits

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why a Section on Special Sequences?

- Theory so far is **abstract** (limits, Cauchy, lim sup)



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- **Goal:** compute limits of five recurring sequences once and for all



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Why a Section on Special Sequences?

- Theory so far is **abstract** (limits, Cauchy, lim sup)
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- **Goal**: compute limits of five recurring sequences once and for all
- **One technique** powers every proof: a simple *squeeze*



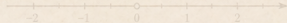
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$$a^2 + b^2 = c^2$$

The Squeeze Principle



$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



A Useful Bounding Remark

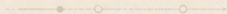
Squeeze to zero

If $0 \leq x_n \leq s_n$ for all $n \geq N$, and $s_n \rightarrow 0$, then $x_n \rightarrow 0$.

Why it works: given $\varepsilon > 0$, eventually $s_n < \varepsilon$, and so $0 \leq x_n \leq s_n < \varepsilon$.

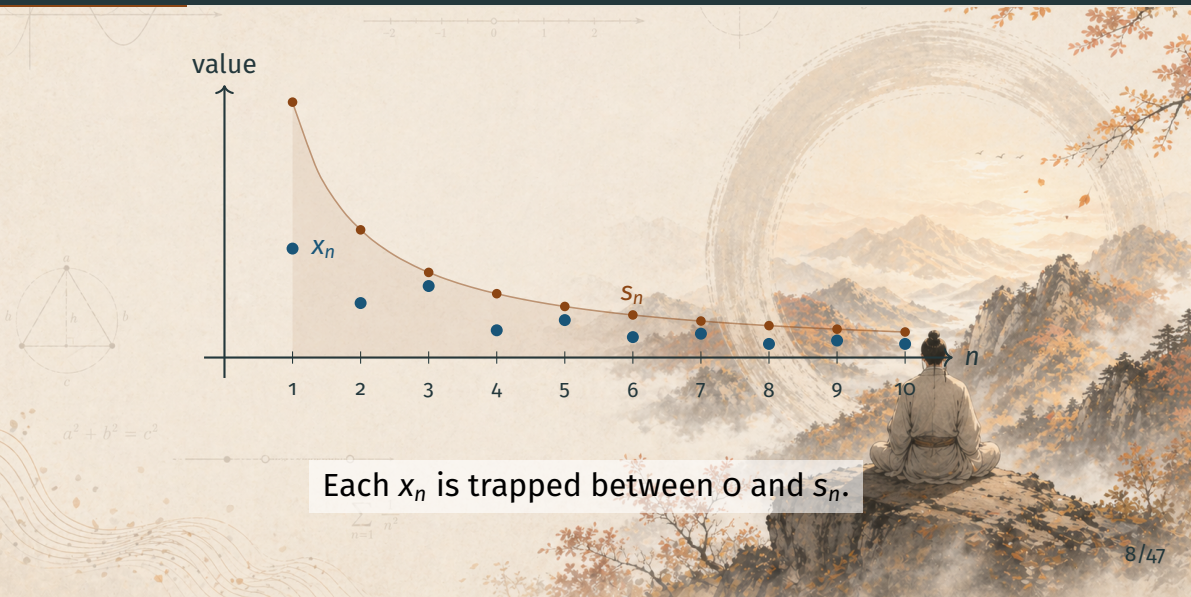


$$a^2 + b^2 = c^2$$

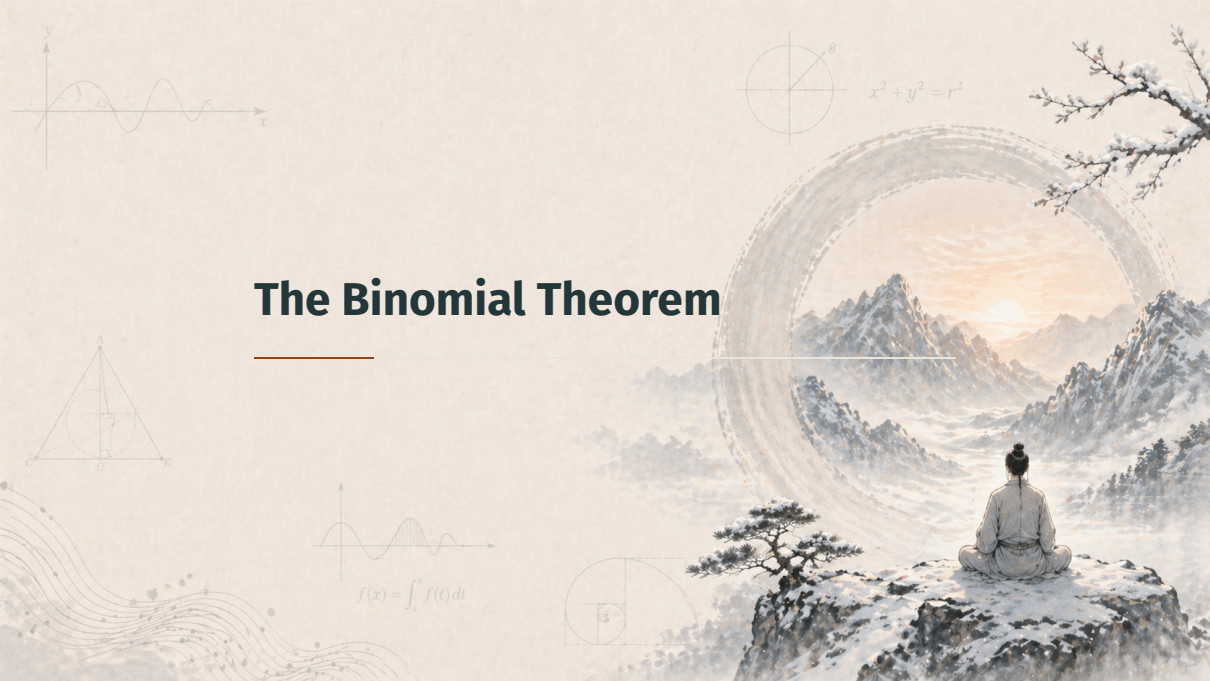


$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Picture: A Vanishing Envelope



The Binomial Theorem



The Factorial

Definition — Factorial

For an integer $n \geq 1$,

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdots 2 \cdot 1, \quad \text{and} \quad 0! = 1.$$

What $n!$ counts

$n!$ is the number of **permutations** of $\{1, 2, \dots, n\}$: the orderings of n distinct objects in a row.

Why: n choices for the first slot, $n - 1$ for the second, $\dots$, 1 for the last.

Example: The $3! = 6$ Permutations of $\{1, 2, 3\}$



From *start*: 3 ways to pick the first entry, then 2, then 1.
Thus $3! = 3 \cdot 2 \cdot 1 = 6$ orderings are the leaves.

The Binomial Coefficient

Definition — Binomial Coefficient

For integers $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n!}{k!(n-k)!}.$$

What $\binom{n}{k}$ counts

$\binom{n}{k}$ is the number of k -element subsets of $\{1, \dots, n\}$.

Why $\binom{n}{k}$ Counts k -Element Subsets

Strategy: count the ordered k -tuples of *distinct* elements of $\{1, \dots, n\}$ in two ways. Let C be the number of k -element subsets.

Two counts of the same collection

1. **Directly:** n choices for the first entry, $n - 1$ for the second, ..., $n - k + 1$ for the last: $n(n - 1) \cdots (n - k + 1) = \frac{n!}{(n - k)!}$ tuples.
2. **By grouping:** each k -element subset can be ordered in $k!$ ways, giving $C \cdot k!$ tuples.
3. **Equate:** $C \cdot k! = \frac{n!}{(n - k)!}$, so $C = \frac{n!}{k! (n - k)!} = \binom{n}{k}$. $\square$

Example: $\binom{4}{2}$ Counts the 2-Subsets of $\{1, 2, 3, 4\}$

Count the ordered pairs of distinct elements two ways:

Directly, $4 \cdot 3 = 12$; or as $(\# \text{ subsets}) \times 2!$.

$\{1, 2\}$

$(1, 2) (2, 1)$

$\{1, 3\}$

$(1, 3) (3, 1)$

$\{1, 4\}$

$(1, 4) (4, 1)$

$\{2, 3\}$

$(2, 3) (3, 2)$

$\{2, 4\}$

$(2, 4) (4, 2)$

$\{3, 4\}$

$(3, 4) (4, 3)$

Each subset accounts for $2! = 2$ ordered pairs, so $\binom{4}{2} = \frac{12}{2!} = 6$.

The Binomial Theorem

Binomial Theorem

For all real a, b and every integer $n \geq 0$,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

We take $0^0 = 1$ for notational simplicity (in general, 0^0 is undefined).

Pascal's Rule — A Counting Proof

Pascal's Rule

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}, \quad 1 \leq k \leq n.$$

Proof by counting

$\binom{n+1}{k}$ is the number of k -element subsets of $\{1, \dots, n+1\}$. Split these subsets by whether or not they contain the element $n+1$:

- **without** $n+1$: a k -subset of $\{1, \dots, n\}$, so $\binom{n}{k}$ of them;
- **with** $n+1$: the element $n+1$ adjoined to a $(k-1)$ -subset of $\{1, \dots, n\}$, so $\binom{n}{k-1}$ of them.

The two cases are disjoint and exhaustive, so $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$. $\square$

The Binomial Theorem — Proof by Induction

Induction on n . Base ($n = 0$): $(a + b)^0 = 1$. Step: assume it holds for n .

Steps

1. **Multiply** $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$ **by** $(a + b)$:

$$(a + b)^{n+1} = \left[\sum_{k=0}^n \binom{n}{k} a^{n+1-k} b^k \right] + \left[\sum_{k=0}^n \binom{n}{k} a^{n-k} b^{k+1} \right].$$

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Reindex the right sum by $k \mapsto k - 1$ (now $k = 1, \dots, n + 1$).

$$(a + b)^{n+1} = \left[\sum_{k=0}^n \binom{n}{k} a^{n+1-k} b^k \right] + \left[\sum_{k=1}^{n+1} \binom{n}{k-1} a^{n+1-k} b^k \right].$$

2. Endpoint terms

- $k = 0$ (left sum): $\binom{n}{0} a^{n+1} = \binom{n+1}{0} a^{n+1}$, since $\binom{n}{0} = \binom{n+1}{0} = 1$.
- $k = n + 1$ (right sum): $\binom{n}{n} b^{n+1} = \binom{n+1}{n+1} b^{n+1}$, since $\binom{n}{n} = \binom{n+1}{n+1} = 1$.

The Binomial Theorem — Proof by Induction

Induction on n . Base ($n = 0$): $(a + b)^0 = 1$. Step: assume it holds for n .

Steps

1. **Multiply** $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$ **by** $(a + b)$:
$$(a + b)^{n+1} = \left[\sum_{k=0}^n \binom{n}{k} a^{n+1-k} b^k \right] + \left[\sum_{k=1}^{n+1} \binom{n}{k-1} a^{n+1-k} b^k \right].$$
2. **Endpoint terms:** $\binom{n}{0} a^{n+1} = \binom{n+1}{0} a^{n+1}$. $\binom{n}{n} b^{n+1} = \binom{n+1}{n+1} b^{n+1}$.
3. **Middle terms** ($1 \leq k \leq n$): both sums contribute, so the coefficient at $a^{n+1-k} b^k$ is $\binom{n}{k} + \binom{n}{k-1}$, which equals $\binom{n+1}{k}$ by **Pascal's rule**.
4. **Together** $(a + b)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k$. $\square$

$$f(x) = \int_a^x f(t) dt$$

Remark: A Combinatorial Proof

Write $(a + b)^n = (a + b)(a + b) \cdots (a + b)$. Expanding the product, each term picks one letter, a or b , from every factor.

(~~a~~+b)

factor 1

(a+~~b~~)

factor 2

(~~a~~+b)

factor 3

(a+~~b~~)

factor 4

(a+~~b~~)

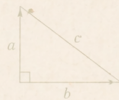
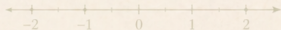
factor 5

$$\Rightarrow a \cdot b \cdot a \cdot b \cdot b = a^2 b^3 \text{ (} b \text{ from factors } \{2, 4, 5\})$$

Why the coefficient of $a^{n-k}b^k$ is $\binom{n}{k}$

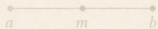
In the picture b comes from the 3-subset $\{2, 4, 5\}$, giving the term $a^2 b^3$.

All $\binom{5}{3} = 10$ of the 3-subsets give $a^2 b^3$. In general, b from a k -subset of $\{1, \dots, n\}$ gives $a^{n-k} b^k$, so $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$.



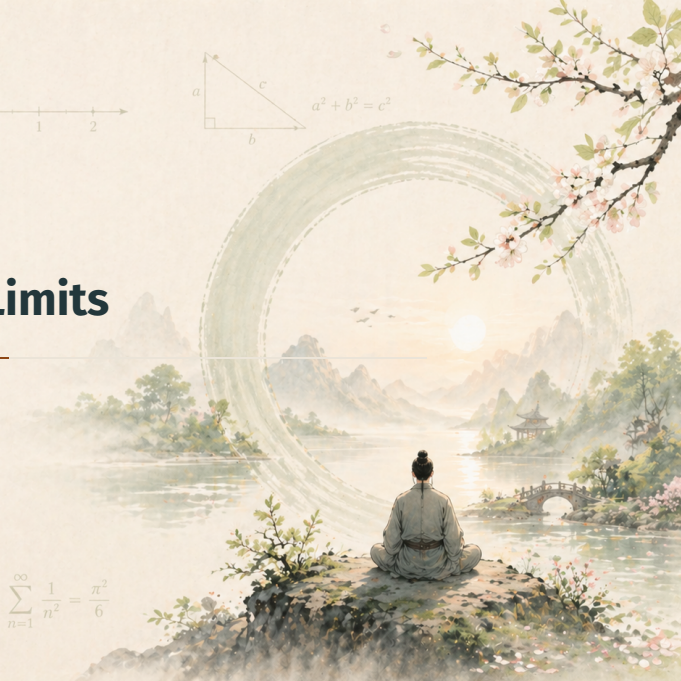
$$a^2 + b^2 = c^2$$

Five Special Limits



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

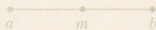
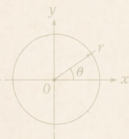
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.20: Five Special Limits

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(a) If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.



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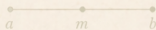
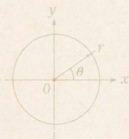
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- (a) If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.
- (b) If $p > 0$, then $\lim_{n \rightarrow \infty} \sqrt[n]{p} = 1$.



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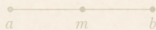
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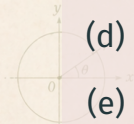
(d) If $p > 0$ and $\alpha \in \mathbb{R}$, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0$.

Theorem 3.20: Five Special Limits

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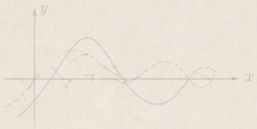
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- (c) $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.
- (d) If $p > 0$ and $\alpha \in \mathbb{R}$, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0$.
- (e) If $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$.


$$a \quad m \quad b$$


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

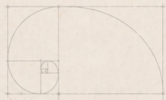


$$e^{i\theta} = \cos \theta + i \sin \theta$$



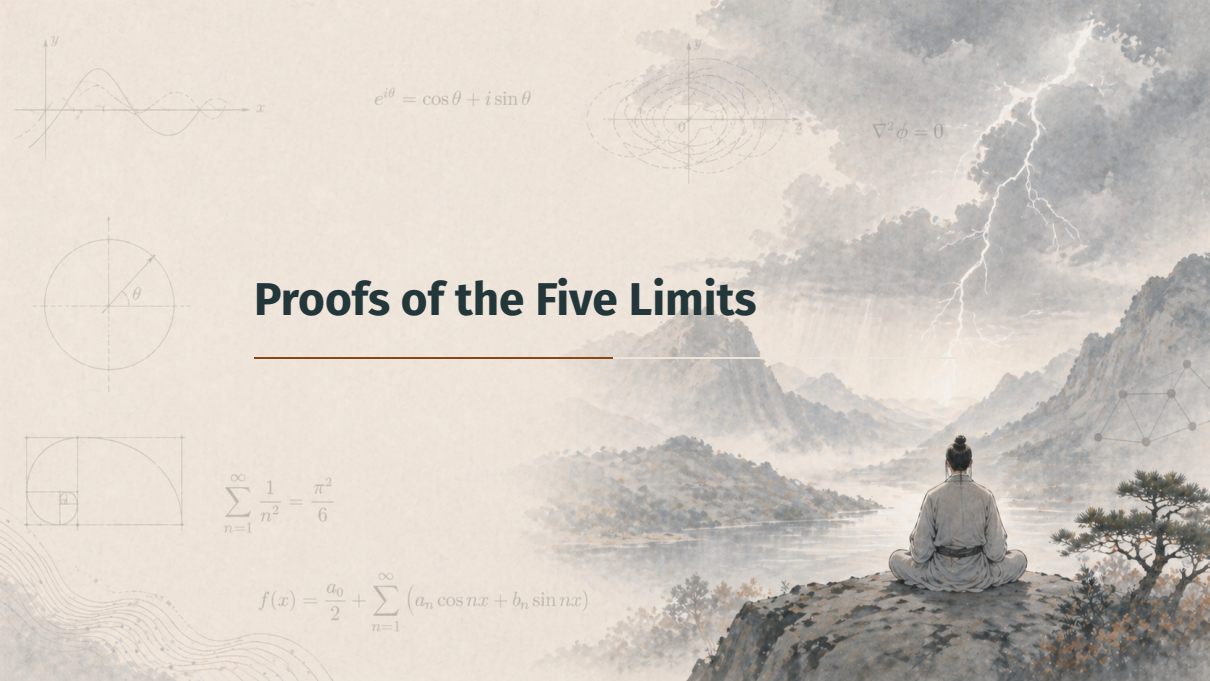
$$\nabla^2 \phi = 0$$

Proofs of the Five Limits



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



Proof of (a): $1/n^p \rightarrow 0$

Step 1: Pick a threshold using the archimedean property

Fix $\varepsilon > 0$. Choose n large enough that

$$n > \left(\frac{1}{\varepsilon}\right)^{1/p}.$$

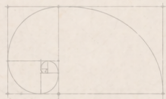
Proof of (a): $1/n^p \rightarrow 0$

Step 1: Pick a threshold using the archimedean property

Fix $\varepsilon > 0$. Choose n large enough that $n > (1/\varepsilon)^{1/p}$.

Step 2: Raise to the p -th power and rearrange

Then $n^p > 1/\varepsilon$, so $\frac{1}{n^p} < \varepsilon$. $\square$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of (b): $\sqrt[n]{p} \rightarrow 1$ — Case $p > 1$

Setup. Put $x_n = \sqrt[n]{p} - 1$. Since $p > 1$, $x_n > 0$ and $\sqrt[n]{p} = 1 + x_n$.

Steps

1. **Binomial bound.** $p = (1 + x_n)^n \geq 1 + nx_n$.

Why $(1 + x)^n \geq 1 + nx$ for $x \geq 0$ (Bernoulli's inequality)

The binomial theorem gives $(1 + x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + x^n$, a sum of nonnegative terms when $x \geq 0$. Dropping all but the first two terms leaves the stated bound.

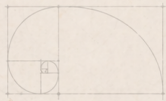
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

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Steps

1. **Binomial bound.** $p = (1 + x_n)^n \geq 1 + nx_n$.
2. **Solve for x_n .** $0 < x_n \leq \frac{p - 1}{n}$.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

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Steps

1. **Binomial bound.** $p = (1 + x_n)^n \geq 1 + nx_n$.

2. **Solve for x_n .** $0 < x_n \leq \frac{p-1}{n}$.

3. **Squeeze.** $\frac{p-1}{n} \rightarrow 0$, so $x_n \rightarrow 0$.

Hence $\sqrt[n]{p} = 1 + x_n \rightarrow 1$. $\square$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of (b): $\sqrt[n]{p} \rightarrow 1$ — Remaining Cases

Case $p = 1$

$\sqrt[n]{1} = 1$ for all n . Trivial.

Case $0 < p < 1$: take reciprocals

Let $q = 1/p > 1$. By the previous case, $\sqrt[n]{q} \rightarrow 1$, so

$$\sqrt[n]{p} = \frac{1}{\sqrt[n]{q}} \rightarrow \frac{1}{1} = 1. \quad \square$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of (c): $\sqrt[n]{n} \rightarrow 1$

Setup. Put $x_n = \sqrt[n]{n} - 1$. Since $\sqrt[n]{n} \geq 1$, $x_n \geq 0$ and $\sqrt[n]{n} = 1 + x_n$.

Steps (for $n \geq 2$)

1. **Binomial, keep the quadratic term.**

$$n = (1 + x_n)^n \geq \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} x_n^2.$$

Why $(1 + x)^n \geq \binom{n}{2} x^2$ for $x \geq 0$ (from the binomial theorem)

The binomial theorem gives $(1 + x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + x^n$, a sum of nonnegative terms when $x \geq 0$. Dropping all but the $\binom{n}{2}x^2$ term leaves the stated bound.

Proof of (c): $\sqrt[n]{n} \rightarrow 1$

Setup. Put $x_n = \sqrt[n]{n} - 1$. Since $\sqrt[n]{n} \geq 1$, $x_n \geq 0$ and $\sqrt[n]{n} = 1 + x_n$.

Steps (for $n \geq 2$)

1. **Binomial, keep the quadratic term.** $n \geq \frac{n(n-1)}{2} x_n^2$.
2. **Solve for x_n .** $x_n^2 \leq \frac{2}{n-1}$, so $0 \leq x_n \leq \sqrt{\frac{2}{n-1}}$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of (c): $\sqrt[n]{n} \rightarrow 1$

Setup. Put $x_n = \sqrt[n]{n} - 1$. Since $\sqrt[n]{n} \geq 1$, $x_n \geq 0$ and $\sqrt[n]{n} = 1 + x_n$.

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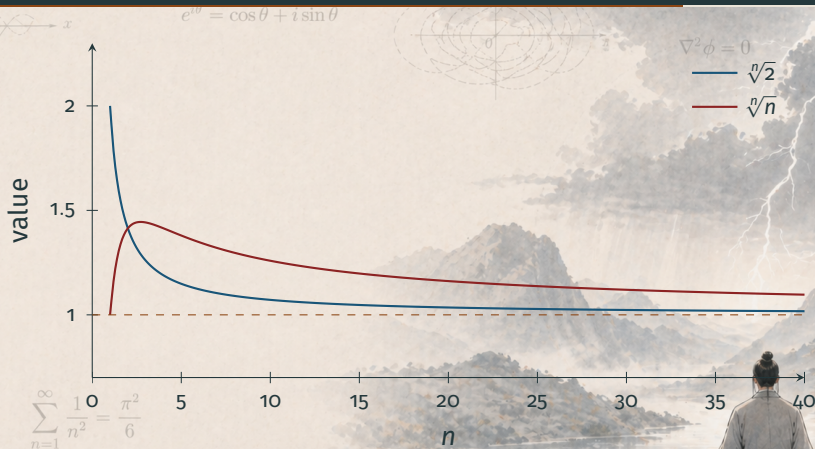
2. **Solve for x_n .** $x_n^2 \leq \frac{2}{n-1}$, so $0 \leq x_n \leq \sqrt{\frac{2}{n-1}}$.

3. **Squeeze.** $\frac{2}{n-1} \rightarrow 0$ by (a), so $\sqrt{\frac{2}{n-1}} \rightarrow 0$.

Hence $x_n \rightarrow 0$ and $\sqrt[n]{n} = 1 + x_n \rightarrow 1$. $\square$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Picture: $\sqrt[n]{p}$ and $\sqrt[n]{n}$ Both Land at 1



Both sequences descend gently toward 1 as n grows.

Proof of (d): $n^\alpha / (1 + p)^n \rightarrow 0$

Setup. Fix an integer k with $k > \alpha$ and $k > 0$.

Idea: bound $(1 + p)^n$ below by a power of n exceeding α .

Steps (for $n > 2k$)

1. **Binomial lower bound.**

$$(1 + p)^n \geq \binom{n}{k} p^k = \frac{n(n-1) \cdots (n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}.$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of (d): $n^\alpha / (1 + p)^n \rightarrow 0$

Setup. Fix an integer k with $k > \alpha$ and $k > 0$.

Steps (for $n > 2k$)

1. **Binomial lower bound.** $(1 + p)^n > \frac{n^k p^k}{2^k k!}$.
2. **Rearrange the ratio.** $0 < \frac{n^\alpha}{(1 + p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}$.

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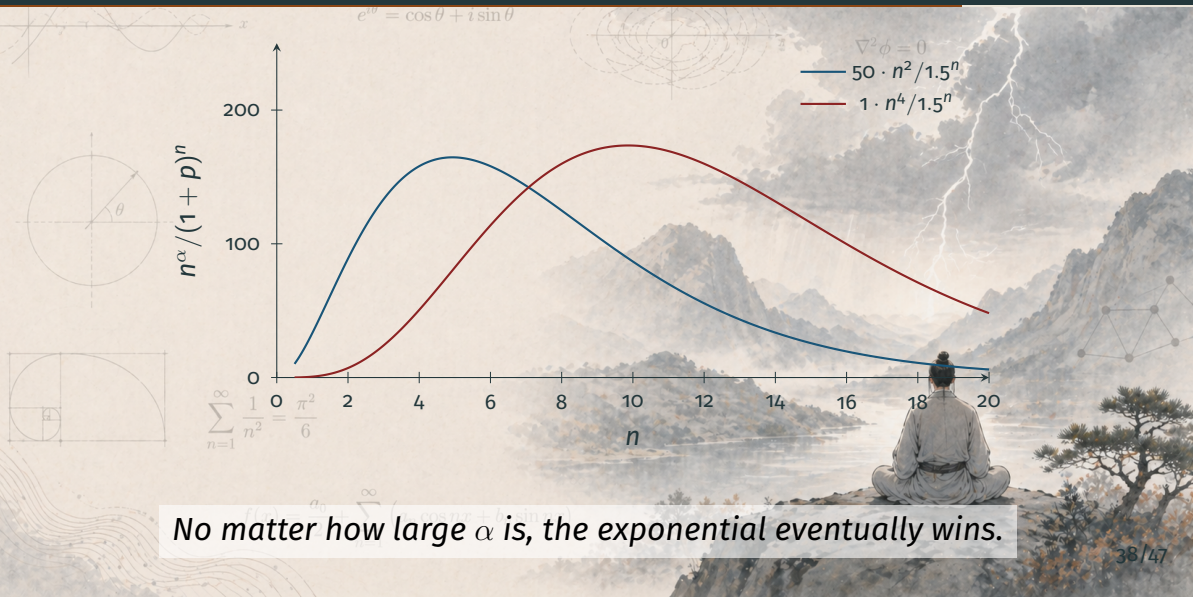
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1. **Binomial lower bound.** $(1 + p)^n > \frac{n^k p^k}{2^k k!}$.
2. **Rearrange the ratio.** $0 < \frac{n^\alpha}{(1 + p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}$.
3. **Apply (a).** Write $\alpha - k = -q$ with $q > 0$.
Then $n^{\alpha-k} = \frac{1}{n^q} \rightarrow 0$, so $\frac{n^\alpha}{(1 + p)^n} \rightarrow 0$. $\square$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Picture: Exponentials Beat Polynomials



No matter how large α is, the exponential eventually wins.

Proof of (e): $x^n \rightarrow 0$ when $|x| < 1$

Setup: write $|x| = 1/(1+p)$

Let x with $|x| < 1$. If $x = 0$, trivially $x^n \rightarrow 0$. Otherwise put

$$p = \frac{1}{|x|} - 1 > 0, \quad \text{so} \quad |x| = \frac{1}{1+p}.$$

Proof of (e): $x^n \rightarrow 0$ when $|x| < 1$

Apply part (d) with $\alpha = 0$

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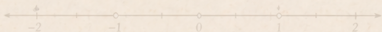
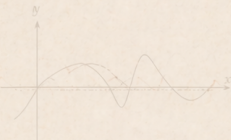
$$p = \frac{1}{|x|} - 1 > 0, \quad \text{so} \quad |x| = \frac{1}{1+p}.$$

$$|x^n| = |x|^n = \frac{1}{(1+p)^n} = \frac{n^0}{(1+p)^n} \rightarrow 0.$$

Since $|x^n| \rightarrow 0$, we have $x^n \rightarrow 0$. $\square$

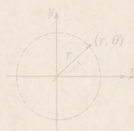
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$$a^2 + b^2 = c^2$$

Summary



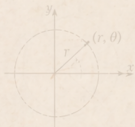
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Key takeaways from this section:

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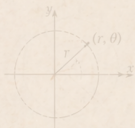
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- **Thm. 3.20 (d):** *exponentials beat polynomials:* $\frac{n^\alpha}{(1+p)^n} \rightarrow 0$
- **Thm. 3.20 (e):** $x^n \rightarrow 0$ for $|x| < 1$ (geometric decay)

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Coming Up Next

Next section: Series

- Partial sums and what it means for a series to converge
- Tests for convergence (Cauchy criterion, comparison, root, ratio)
- Many of these tests will lean on the five limits we just established

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

